

Docker Final Project Assignment

Project Title

Containerized Guestbook Application

Objective

Build, containerize, secure, and deploy a multi-container Guestbook application using Docker.

Students must demonstrate understanding of:

- Docker Images
- Dockerfile
- Docker Compose
- Docker Networks
- Docker Volumes
- Docker Hub
- Docker Scout
- GitHub

Scenario

Your company wants to deploy a Guestbook application where users can submit messages.

The application consists of:

1. Frontend (Python Flask Web App)
2. Database (Redis)

Both services must communicate using Docker Compose.

Tasks

Part 1: Containerization

Create a Dockerfile for the Flask application.

Requirements:

- Use Python base image
- Install dependencies

- Copy application code
- Expose application port
- Start application

Part 2: Docker Compose

Create a docker-compose.yml file that:

- Builds the Flask application
- Runs Redis
- Creates a custom network
- Uses a named volume for Redis persistence

Part 3: Docker Networking

Verify communication between:

- Flask Container
- Redis Container

The application must use the Redis service name as the hostname.

Part 4: Docker Volumes

Configure Redis to use a persistent Docker volume.

Verify that data persists after:

`docker compose down`

`docker compose up -d`

Part 5: Docker Scout

Run vulnerability analysis on your image.

Commands:

`docker scout quickview`

`docker scout cves`

Document findings in README.md.

Part 6: Docker Hub

Push your application image to Docker Hub.

Required:

docker login

docker tag IMAGE_NAME USERNAME/guestbook:v1

docker push USERNAME/guestbook:v1

Part 7: Documentation

Create README.md containing:

- Project Overview
- Architecture Diagram
- Build Steps
- Run Steps
- Docker Compose Explanation
- Docker Network Explanation
- Docker Volume Explanation
- Docker Scout Results
- Docker Hub Image Link

Submission

Submit:

1. GitHub Repository Link
2. Docker Hub Image Link